

# HQOD: Harmonious Quantization for Object Detection

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**Abstract**—Task inharmony problem commonly occurs in modern object detectors, leading to inconsistent qualities between classification and regression tasks. The predicted boxes with high classification scores but poor localization positions will worsen the performance of detectors after Non-Maximum Suppression. Furthermore, when object detectors collaborate with Quantization-Aware Training (QAT), we observe that the task inharmony problem will be further exacerbated, which is considered one of the main causes of the performance degradation of quantized detectors. To tackle this issue, we propose the Harmonious Quantization for Object Detection (HQOD) framework, which consists of two components. Firstly, we propose a task-correlated loss to encourage detectors to focus on improving samples with lower task harmony quality during QAT. Secondly, a harmonious Intersection over Union (IoU) loss is incorporated to balance the optimization of the regression branch across different IoU levels. The proposed HQOD can be easily integrated into different QAT algorithms and detectors. Remarkably, on the MS COCO dataset, our 4-bit ATSS with ResNet-50 backbone achieves a state-of-the-art mAP of 39.6%, even surpassing the full-precision one. Codes are available at <https://github.com/Menace-Dragon/VP-QOD>.

**Index Terms**—Object Detection, Task Inharmony, Model Quantization, Quantization-Aware Training

## I. INTRODUCTION

Deep Convolutional Neural Networks (CNNs) have made remarkable strides in various applications of object detection [1]–[4]. Nevertheless, these detectors with outstanding performance exhibit significant computational and parameter demands, facing challenges to efficiently run on devices with limited resources (e.g., mobile phones or drones).

In modern object detectors, the Task Inharmony (TI) problem has always been a focal point of attention [5]–[7]. Usually, a multi-task pipeline is used to generate both location coordinates and corresponding labels for an object, including a classification branch and a regression branch with two parallel heads. This parallel design can result in an inconsistent distribution of classification scores and regression scores (i.e., Intersection over Union, IoU). Moreover, to efficiently deploy detectors on resource-constrained devices, the application of model compression techniques is necessary.

As one of the popular model compression techniques, Quantization-Aware Training (QAT) methods [8]–[12] are introduced as the widely accepted approach to achieve low-bit

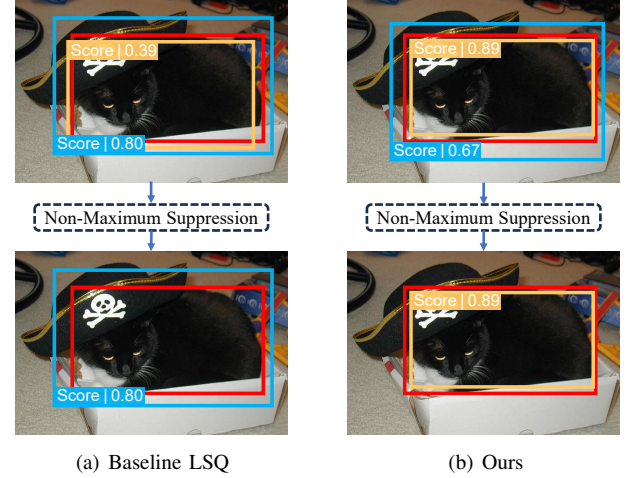


Fig. 1. The results of object detection between the baseline LSQ [8] and our proposed framework. The ground truth is represented by the red bounding box, and the classification scores of the predicted boxes are explicitly indicated. Our framework enhances the harmonious relationship between the classification task and regression task, producing more accurate bboxes.

quantization while preserving near full-precision performance. By simulating the feedforward quantization operations during time-consuming training or fine-tuning, the network can readily adjust to the quantization noise, leading to more optimal solutions. LSQ [8] treats the quantization parameter, step size  $s$  as a learnable parameter, enabling  $s$  to adaptively learn optimal casting. TQT [11] adopts the concept of learnable  $s$  and applies it to hardware-friendly power-of-2 (PoT) scale quantization, achieving exciting results in PoT scale quantization. AQD [12] introduces the Sync-BN in the shared detection head to alleviate the suboptimal issue of  $s$  arising from the multi-level inputs. However, when quantizing low-bit object detectors, the performance still exhibits substantial gaps compared to the full-precision ones.

In this work, we identify that the exacerbation of the TI problem under low-bit constraints is one of the primary reasons leading to the performance gap. Taking LSQ [8] as an example, the quantized RetinaNet [2] with ResNet-18 [13] backbone generates a representative instance of inconsistent bounding boxes, as shown in Fig. 1(a). There are two inharmonic candidates (a yellow bounding box and a blue bounding box) and one ground-truth box colored in

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